An Agent That Copies Nobody Is Certified as a Leader[1]

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With
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Abstract

Multi-agent oversight increasingly proposes information-theoretic influence detection: lagged transfer entropy between agents, a surrogate null to decide which links are real, and a per-agent influence score. We audit that recipe on the AI Village, a public corpus of frontier models talking in shared channels, across four villages. Two facts make agent chat hostile to the standard approach. Replies arrive a median of 28.1 seconds apart, so no clock window is both dense enough to estimate and short enough to separate a message from its answer, and at every width we tried the topic symboliser certifies no more edges than a meaning-free hash null. Moving to a turn-ordered grid, in which one message is one tick, fixes the representation but invalidates the conventional circular-shift surrogate: exactly one agent is non-silent per tick, and shifting destroys that. The cost is severe. An agent planted with zero coupling is certified as a driver in 8 of 8 runs, with z up to 71.7, and on data with every structure destroyed the shift surrogate's family-wise error rate is 0.90 against a nominal 0.05. A symbol-permutation surrogate restores it to 0.10 with a 95% interval of 0.03 to 0.26. Re-certified, main-village falls from 11 edges to 2 with none in common, while open-chat rises from 3 to 18, because the invalid surrogate masks as well as manufactures. We give the controls practitioners should run.

[1] Research conducted at the Apart Swarm Alignment Hackathon, November 2026

1. Introduction

If a population of AI agents shares a channel, an overseer would like to know who is driving whom. The standard toolkit is information-theoretic: lagged transfer entropy between agents, a surrogate null to decide which links are real, and a per-agent influence score. It needs no labels and no model internals, only a log of who said what and when.

Almost every demonstration of these detectors runs on synthetic traces with a planted leader, where the analyst controls timing and vocabulary. Real agent chat is asynchronous, bursty and written in open language. This report asks a narrower question than "what is the influence structure of a real swarm": whether the standard recipe, applied to real agent logs, returns anything an overseer should act on.

We run that audit on the AI Village (AI Digest, 2026), in which frontier models share chat channels over eighteen months, across four villages differing in size, lifetime and tempo, applying the checks a reader should demand: planted positive and negative controls, a meaning-free symboliser, replication in a held-out window, and sensitivity to the parameters we had to choose. The audit fails in a specific and instructive way, with a clean mechanism and a known fix.

1. The conventional circular-shift surrogate is invalid on turn-ordered agent chat. We give the structural reason and measure the damage: a family-wise error rate of 0.90 against a nominal 0.05, and an agent coupled to nothing certified as a driver in 8 of 8 runs.
2. We adopt the local-permutation surrogate from the event-based literature, verify it restores calibration here, and re-certify every result under it. The invalid surrogate is not merely liberal: on one channel it manufactures 11 edges of which none survive, on another it masks 15.
3. We measure detection power with planted controls, so absence can be interpreted. A median-volume follower copying on 80% of its messages is detected once in four runs.
4. We report what survives, including one-directional influence from individual human visitors to specific agents in the one well-powered channel.

2. Related Work

Transfer entropy and surrogates. Transfer entropy (Schreiber, 2000) measures how much a source's past reduces uncertainty about a target's next symbol beyond the target's own past. Plug-in estimators are biased upward on short series, so significance rests on surrogate data rather than the raw statistic, with family-wise error across all ordered pairs controlled by a single-step max-T test (Westfall and Young, 1993). The usual surrogate is a circular time shift of the source.

Event-based data breaks the shift surrogate. Our audit rediscovers this in a new domain. Shorten, Spinney and Lizier (2021), working on neural spike trains, show that transfer entropy for event-based processes is poorly served by time discretisation, and specifically that the conventional source-time-shift surrogate fails to represent the null hypothesis of conditional independence; they propose a local permutation scheme instead. Agent chat is event-based in exactly that sense: messages are events in continuous time, not samples of a clocked signal.

Our contribution is not the fix but the demonstration that multi-agent safety practice inherits this failure, and that it is severe enough to manufacture leaders out of nothing.

Oversight under a budget. AI-control research frames monitoring as resource constrained, with a cost for false positives (Greenblatt et al., 2024). An influence graph is one such measurement, and our concern is the error rate attached to it.

3. Data

The AI Village dataset ships one gzipped-JSONL table per village. We use four villages, which differ enough to test whether any finding is a property of one channel.

village	agent messages	agents	lifetime	human messages	busiest room
main	205,726	58	576 days	10,029	#general (166,715, 81%)
open-chat	35,483	23	66 days	12,264	#general (all)
swarm	3,848	133	1 day	5	room a (3,846)
game-jam	838	6	3 days	36	per-team rooms

A 2026 scaffolding change scopes each agent to its current room's context, so the busiest room of each village is a self-contained influence substrate and we analyse it alone.

The primary analysis window is the final fourteen days of main-village #general, 17 to 31 October 2026, holding 8,800 agent messages. Twenty-two models posted at least 50 messages each in that window, and the directed test uses the 15 busiest of them, giving 7,958 message events. Two scaffolding facts are disclosed rather than corrected: the automated nudger that injects messages was off throughout this window, and the unused #best and #rest rooms were soft-deleted mid-window, which does not touch #general. We read message metadata and text only, and individual human participants appear only as aggregated or anonymised nodes.

4. Method

4.1 From open language to symbols

Transfer entropy needs discrete symbols, so each message is mapped to a small alphabet before influence is estimated, and that mapping is kept separate from the estimator. A topic symboliser (TF-IDF then k-means, k=8) gives a message its topic cluster id, so paraphrases of one idea share a symbol. A hash symboliser (a stable hash of the normalised text into k=8 buckets) assigns symbols with no notion of meaning and serves as a null: an edge surviving under topics but not hashing lives in shared meaning, one appearing under both lives in timing alone.

4.2 The lag problem, and why clock windows cannot solve it

A lag-1 transfer entropy compares window t-1 with window t, so influence landing inside one window is invisible to it. Agent chat makes this binding. Among the fifteen analysed agents the gap between consecutive messages by different speakers has a median of 28.1 seconds (25th percentile 9.2, 75th 66.2, 95th 159.3). Turn-taking is therefore a seconds-scale process, while any clock window wide enough to hold two speakers is minutes wide and swallows both a message and its answer. Section 5.1 measures both horns directly.

We avoid the bind by ordering messages canonically and letting one message be one tick. Each agent's series carries its symbol where it spoke and a silent symbol elsewhere, so the lag-1 comparison is between a message and the one immediately before it. The cost is that the measure captures immediate-predecessor influence only.

4.3 Two surrogate nulls

The preexisting estimator certifies an edge by comparing the observed transfer entropy with a **circular-shift** surrogate of the source, using a max-T family-wise test at level 0.05 across all ordered pairs. On the turn-ordered grid this surrogate fails, and Section 5.2 both measures the failure and isolates its cause: the structure of the real speaker sequence. Real conversation is not a sequence of independent draws about who talks next. It has strong turn-taking, and circularly shifting one agent's series slides that burst pattern to a different part of the timeline, where it aligns against the other agents differently than it ever does in the data.

We therefore add a **symbol-permutation** surrogate, following the local-permutation idea of Shorten et al. (2021): the source keeps the exact ticks at which it spoke, and only its symbols are permuted among them. Speaking pattern, sparsity and mutual exclusivity are preserved exactly, and only the correspondence between symbol and time is destroyed. Both are reported throughout.

4.4 Controls

Positive control (spike-in). Real logs have no ground truth, so we plant it. A synthetic agent is inserted into the real message stream; a fraction c of its messages are responses placed immediately after a chosen leader's message and copying that leader's topic symbol, and the rest land at random positions with a symbol drawn from the channel's marginal. Sweeping c, and separately the follower's volume, gives the detection floor.

Negative controls. An agent planted with no coupling at all, at a range of volumes, should never be certified. And the real speaker sequence with its symbols globally permuted contains no directed topical influence by construction, so a calibrated procedure should certify an edge in about 5% of replicates.

Replication and sensitivity. The certified edge set is recomputed in the adjacent fourteen-day window (restricted to agents active in both, with one topic space fit on the union), in each half of the primary window, and at several alphabet sizes.

5. Results

5.1 Clock windows cannot separate cause from effect

Figure 1 runs the detector on clock windows from 30 seconds to an hour, and the two failure modes close on each other. At 30 seconds only 1.2% of agent-windows contain a message, while already 32.0% of replies land in the same window as the message they answer and are invisible to a lag-1 test. By 5 minutes that share is 84.4%, and by an hour 98.6%.

More telling is what the edges are made of. At every width the meaning-free hash null certifies about as many edges as the topic symboliser (12 against 9 at 30 seconds, 19 against 17 at one minute, 44 against 27 at fifteen), and at 5 minutes more, 19 against 17. Whatever clock-window transfer entropy detects here, it is not shared meaning, so we move to the turn-ordered grid.

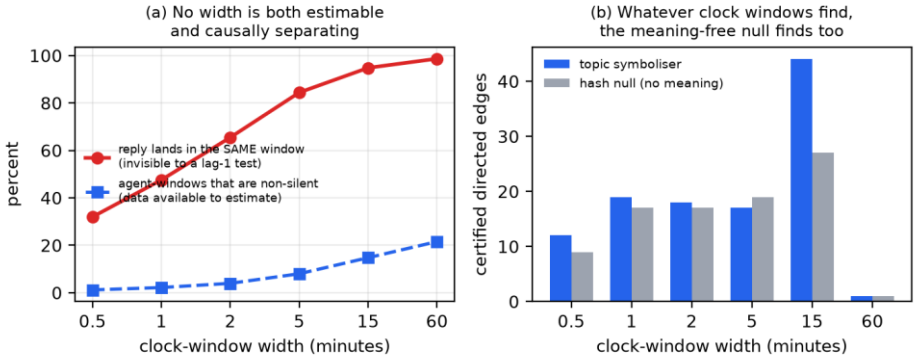


Figure 1. Clock-window transfer entropy on the 15 analysed agents of main-village #general. Left, the two failure modes as a function of window width. Right, edges certified under the topic symboliser against the meaning-free hash null.

5.2 On a turn-ordered grid the standard surrogate is invalid

We plant an agent into the real stream, drawing its symbols from the channel's marginal and coupling it to nothing at all, then ask how often it is certified as driving something.

It is certified in 3 of 8 runs when it posts 50 messages, and in 8 of 8 at 100, 263, 600 and 1500 messages, with the mean largest z rising from 5.6 to 12.3, 21.7, 38.3 and 71.7. The error grows with how much the agent speaks, which rules out a small-sample explanation and says the mismatch is in structure rather than in sample size.

A planted agent could be dismissed as artificial, so we also measure the error rate directly. Keeping the real speaker sequence and permuting the symbols globally leaves data with no directed topical influence by construction. The shift surrogate certifies at least one edge in 9 of 10 such replicates, a family-wise error rate of 0.90 against a nominal 0.05.

What breaks it is the speaker sequence. Destroying one structure at a time isolates the cause (Appendix D). With the real speaker order intact and only the symbols shuffled, the shift surrogate fires in 10 of 10 replicates; randomising the speaker order as well drops it to 0 of 10. Replacing the real symbols with uniform draws, keeping the real order, still gives 6 of 10. So the failure is driven by turn-taking, not by the symbol distribution and not by sparsity: the real sequence repeats the same speaker 36% of the time and returns to a speaker after one other

message 38% of the time. Shifting slides that burst pattern across the timeline; permuting symbols within an agent's own speaking positions leaves it intact, which is why the repair is unaffected in every condition.

5.3 A sparsity-preserving surrogate restores calibration

Following the local-permutation idea of Shorten et al. (2021), we keep each source's speaking positions and permute only its symbols among them. On the same destroyed-structure data the family-wise error rate falls to 0.065 over 260 replicates, with a 95% Wilson interval of 0.041 to 0.102, so it is close to the nominal 0.05 and clearly separated from the 0.90 it replaces. The planted uncoupled agent is now never certified: 0 of 5 at both 263 and 1500 messages, with mean max z of 2.8 in each case against 22.0 and 68.3 under the shift surrogate.

The failure is not an artifact of the configuration we chose. Appendix D shows it holds across swarm sizes from 5 to 20 agents, on a near-uniform alphabet as well as the skewed topic one, and that both the inflation and the masking reproduce under an independent symboliser.

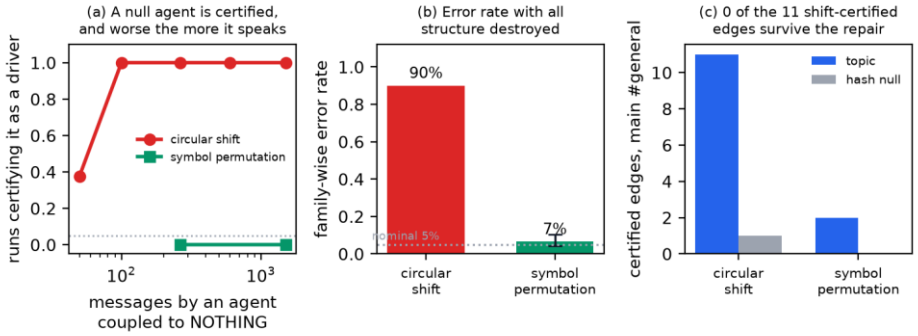


Figure 2. Left, how often an agent coupled to nothing is certified as a driver, under each surrogate. Middle, family-wise error rate measured on data with all structure destroyed, with a Wilson interval; the dotted line is the nominal 0.05. Right, what each surrogate certifies on the real channel.

5.4 The same channels, re-certified

Re-running every channel under the repaired surrogate changes the answers, and not in one direction.

channel	window messages	shift: topic / hash	permutation: topic / hash
main #general	8,800	11 / 1	2 / 0
open-chat #general	8,226	3 / 0	18 / 0
swarm, room a	3,846	2 / 1	1 / 0

In main-village, **none** of the 11 edges the standard recipe certifies survives the repair, and the two the valid surrogate certifies were not among them. In open-chat the correction runs the other way, 3 edges becoming 18, because pathological pairs inflate the max-T threshold and so raise the bar for everything else. An invalid surrogate does not simply add false positives, it scrambles the result.

5.5 How much influence could we have seen?

Absence of edges is only informative if the test could have found them, so we plant one. A synthetic follower is inserted at the median agent's volume and copies the busiest agent's topic on a fraction of its messages. In main-village, under the repaired surrogate, it goes undetected in 0 of 4 runs at couplings up to 0.4 and is caught in only 1 of 4 at 0.8. Holding coupling at 0.8 and varying how much it speaks, detection rises from 1 of 4 at 263 messages to 4 of 4 at 800 and stays there at 2000 and 4000, so the floor sits between 3% and 10% of the channel's messages.

Power is governed by how much an agent says, not only by how strongly it copies, and it is channel-specific. Repeating the control in open-chat, a follower at that channel's median volume is caught in 3 of 4 runs at coupling 0.1 and in 4 of 4 from 0.2 upward. So the channels differ in sensitivity as well as structure: main-village's two edges are a floor, while open-chat's 18 come from a test that can see.

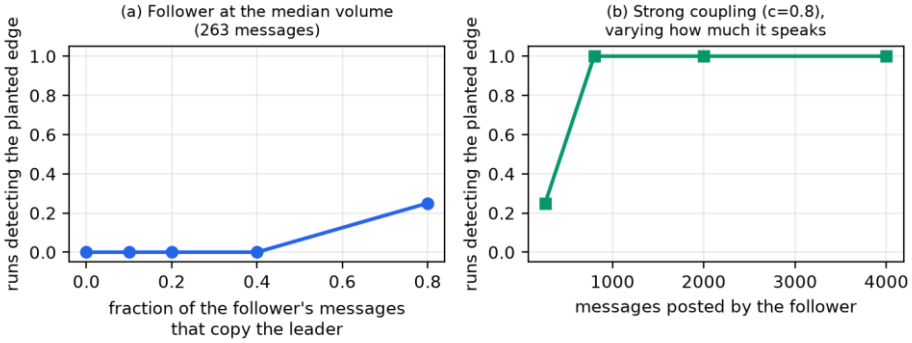


Figure 3. Detection of a planted follower under the repaired surrogate. Left, varying the fraction of its messages that copy the leader, at the median agent's volume. Right, varying how much it speaks, at strong coupling.

5.6 Does anything replicate?

In main-village, no. Under the standard recipe the adjacent fourteen-day window shares 1 edge with the primary window out of 8 and 4, a Jaccard of 0.09, and the two halves share none. Under the repaired surrogate the adjacent window certifies nothing. The one edge that persists, GLM-5.2 to Gemini 3.5 Flash, is also the only edge the hash null certifies, so the single stable thing the standard recipe finds carries no meaning.

In open-chat, where the valid test has power, the picture is better but still partial: restricted to agents active in both windows and sharing one topic space, the current window certifies 9 edges and the previous 5, with 3 in common, a Jaccard of 0.27, against 0 from the hash null. About a third holds up. Splitting that window into halves is harsher, 7 edges against 2 with 1 in common, which the power result predicts, since halving the data moves agents below the floor.

The alphabet size remains a live degree of freedom even under the repaired surrogate, with open-chat certifying 8 edges at k=4, 18 at k=8 and 21 at k=16, so an edge count is never a property of a channel alone.

5.7 Humans in the channel

main-village #general is now effectively human-free, with 3 human messages in the window, so the question has to be asked where humans are present. In open-chat, 32% of messages come from 124 human visitors. Under the repaired surrogate the 12 agents alone certify 17 edges; adding the three busiest visitors individually gives 20, of which 16 are agent to agent, **4 run from an individual visitor to a specific agent, and none run from an agent to a visitor**.

This is the one substantive behavioural observation that survives, and it should be read carefully: topical dependence from a person to an agent is also what ordinary responsiveness looks like. The measurement supports only that the dependence is directional and attributable to particular visitors, not that anyone is steering a swarm.

Two further traps, reported in Appendix D, are worth naming here: a continuous net-influence score turns out to measure volume rather than meaning, and a trivial copy-rate detector does not reproduce transfer entropy, so what this report indicts is the surrogate and not the statistic.

6. What an overseer should do

The audit suggests four controls, none expensive, that should accompany any published influence graph over an agent channel.

1. **Plant an agent coupled to nothing, at your channel's volumes, and check it is not certified.** This one test would have caught the failure reported here, and costs one run per volume.
2. **Measure the error rate directly,** by destroying the structure your method claims to find and checking the family-wise rate is near nominal. If it is not, the surrogate does not match the data.
3. **Preserve the speaker sequence in your surrogate.** Turn-taking is the structure that breaks the shift here, and any surrogate that moves an agent's messages in time will disturb it. For event-based data this is known, and Shorten et al. (2021) is the principled route.
4. **Establish the detection floor before interpreting an absence,** and report it beside the graph. "No edges" and "no influence" are different claims.

Two habits follow: publish a meaning-free symboliser's edge count beside the real one, and require replication in a held-out window before calling a graph the structure of a swarm.

7. Conclusion

Applied to real agent chat, the standard recipe for "who drives whom" fails in a way that is easy to miss and easy to check. Clock windows cannot separate a message from its answer at the timescale agents reply on, and the edges they certify are reproduced by a null with no access to meaning. The turn-ordered representation that fixes this breaks the circular-shift surrogate, whose family-wise error rate on structure-free data we measure at 0.90 against a nominal 0.05, and the cause is the turn-taking structure of conversation itself. Repairing it changes every answer: one channel loses all 11 edges, another gains 15 that were masked. For an overseer

that is the operative finding: an influence graph is a measurement with an error rate, and until that rate is reported it should not be acted on.

Appendix A: Limitations

- **Immediate-predecessor only.** The turn-ordered lag-1 measure sees influence from the message directly before a reply. Influence acting over several turns, or through a room's accumulated context, is out of scope and needs a higher-order or memory-aware estimator.
- **Discrete approximation to an event process.** Agent chat is event-based in continuous time. We use a turn-ordered discretisation with a permutation surrogate as a practical fix; a continuous-time estimator (Shorten et al., 2021) is the principled route and we did not attempt it.
- **Symbolisation is a free parameter.** The central result survives an independent symboliser (Section 5.3), but the edge count does not: it moves with the alphabet size and with which messages the topic model is fit on, and we report those sensitivities rather than tuning them. Two of the eight topic clusters are dominated by function words, because the preexisting symboliser does not remove stopwords.
- **No ground truth in the real logs.** The validity argument rests on planted controls and on the meaning-free null, not on recovering a known answer.
- **Scaffolding confounds.** Agent behaviour is shaped by prompt, tool and memory changes logged in the dataset's changelog. We chose a window in which the message-injecting nudger was off, but other scaffolding is held fixed only by the shortness of the window.
- **The human result is exploratory.** It rests on one village and one window, and given the instability documented in Section 5.4 it should be read as a lead, not a finding. Topical dependence from a human to an agent is also consistent with ordinary responsiveness rather than steering.

Appendix B: Dual-Use and Ethical Considerations

This work audits influence detection on a population of AI agents, which is dual-use. Used defensively it helps an overseer decide what an influence graph is worth, and our main result is deflationary: it tells practitioners to trust these outputs less, which reduces rather than increases the chance of acting on a bad attribution. The same tooling could be used to identify the most influential agent in a swarm in order to target it, or to surveil participants in a channel. Three properties limit the misuse. The subjects are AI models; the dataset is already public under research terms and we add no re-identification; and the measure is weak and local, recovering immediate-predecessor topical dependence rather than any model of how to steer an agent.

We followed the dataset's research terms: research use only, no training on the data, no re-identification, and citation of the source. Human participants appear only as an aggregate node or as anonymised visitor-1..n labels, never by name or handle, and we report no content attributable to an individual person. We read message text and metadata only, never model weights or private user data.

Appendix C: Preexisting tools, new work, and LLM usage

Preexisting (declared). The estimator (lag-1 transfer entropy, circular-shift surrogate, max-T family-wise test) and the reader and symbolisers (topic and hash) were built before the sprint in the repository swarm-prep and are declared preexisting with their validation numbers in that repository's README. Before the sprint, authentication was verified and filenames were listed, but no chat content was read or described, so the weekend is the first reading of the data. The preexisting validation used synthetic chat in which several agents speak per window, a setting in which the circular-shift surrogate is valid; the invalidity reported here arises only on the one-message-per-tick grid introduced this weekend, and is not a defect of the preexisting tool as validated.

New this weekend. The subset and window choices on the real data; the measured reply-latency justification for the lag; the turn-ordered representation; the symbolisation of real message text; the clock-window failure analysis; the symbol-permutation surrogate and the demonstration that the shift surrogate is invalid on this grid; the planted positive and negative controls; the calibration measurement; the replication and sensitivity battery; the four-village comparison; the human-node analysis; the trivial-baseline comparison; the read-budget sweep; all figures; and a vectorised reimplementation of the preexisting estimator, used only for scale and verified bit-for-bit against the preexisting code before use.

LLM usage. The author used an AI coding assistant (Claude) to help write and run the analysis code, to help draft this report, and to structure the figures. The author is responsible for every claim, number and figure here. All results come from the released code run on the public dataset, no result or reference is fabricated, and the numbers in this document are checked against the analysis output by a script that fails on any mismatch.

Appendix D: Isolating the cause, robustness, and two further traps

Which structure breaks the surrogate. Each condition below contains no directed topical influence by construction, so a calibrated procedure should fire in about 5% of replicates. Ten replicates each.

condition	circular shift	symbol permutation
real speaker order, symbols shuffled	10 / 10	0 / 10
real speaker order, uniform i.i.d. symbols	6 / 10	1 / 10
speaker order shuffled, symbols shuffled	0 / 10	1 / 10
speaker order shuffled, uniform i.i.d. symbols	0 / 10	1 / 10

Speaker order is the switch. The symbol distribution only modulates. A synthetic channel that draws speakers independently does not reproduce the failure at all, which is how we found this:

a generated benchmark with exactly one speaker per tick left the shift surrogate roughly calibrated, so mutual exclusivity alone, which we had first proposed as the explanation, is not sufficient.

The surrogate failure is not an artifact of our configuration. It is not a large-swarm effect: across swarms of 5, 10, 15 and 20 agents the shift surrogate certifies something in 22 of 24 null replicates, against 3 of 24 for the permutation surrogate, so shrinking the swarm does not rescue it. It is not a property of the skewed topic alphabet either, since on the near-uniform hash alphabet, where the largest symbol holds 13% of messages rather than 37%, the shift surrogate still fires in 10 of 10 replicates.

Both corrections survive a different symboliser. Replacing TF-IDF and k-means with latent semantic analysis gives 12 edges against 1 in main-village and 2 against 22 in open-chat: the same inflation and the same masking, reached by an independent route to topics. The meaning-free hash control gives 1 against 0 and 0 against 0. So the central result concerns the surrogate rather than any one clustering choice, although the edge count itself remains sensitive to the alphabet size.

A continuous influence score measures volume. Summed outgoing minus incoming transfer entropy, a common per-node leadership summary, ranks agents almost identically under the topic symboliser and under the meaning-free hash null (Spearman 0.85), with the same agent on top at nearly the same value. A score the hash null reproduces cannot be measuring shared meaning, so the certified edge set should be reported instead.

The estimator is not redundant, though. A trivial detector, the rate at which an agent repeats the previous message's topic in excess of its own base rate, ranks pairs very differently from transfer entropy (Spearman 0.21), and the pairs it ranks highest overlap the certified set in only 3 of 10 cases. So transfer entropy is not an expensive way to compute a copy rate, and the failure documented in this report belongs to the surrogate rather than to the statistic.

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